

P04. Real-time Cooperative Vehicle Coordination at Unsignalized Road Intersections

中英双语博士精读笔记

Jiping Luo; Tingting Zhang; Rui Hao; Donglin Li; Chunsheng Chen; Zhenyu Na; Qinyu Zhang

2023 · IEEE Transactions on Intelligent Transportation Systems

Theme / 主题: DRL cooperative trajectory optimization

Method family / 方法族: TD3 deep reinforcement learning for coordination / TD3 深度强化学习协同控制

PDF pages / 页数: 15 **Relevance / 相关度:** 91/100

Source / 来源: <https://arxiv.org/abs/2205.01278>

1 论文全景速览与核心贡献 / High-Level Overview & Core Contributions

1.1 研究背景与痛点 / Background & Pain Points

这篇论文位于“DRL cooperative trajectory optimization”方向。对你的 JSSP-based Intersection Management 课题，它回答的是高层调度、低层轨迹平滑、安全过滤或真实验证中的一个关键子问题：如何让车辆在路口少停、少冲突、少能耗，同时保持在线可执行。

The paper belongs to the theme of DRL cooperative trajectory optimization. For a JSSP-based intersection-management thesis, it illuminates one of the key layers: scheduling, smoothing, safety filtering, or real-world validation.

Pain point / 痛点: Centralized control-authority handoff and DRL policy behavior may be difficult to certify in mixed traffic.

1.2 核心科学问题 / Core Scientific Question

能否用连续动作深度强化学习直接学习无信号交叉口的实时协同轨迹控制策略？

Can continuous-control deep reinforcement learning learn real-time cooperative trajectories for unsignalized intersections?

1.3 科学贡献 / Scientific Contributions

- 方法贡献 (method contribution): Formulates unified cooperative trajectory optimization for unsignalized intersections and solves the sequential decision problem with TD3.

- 安全/避碰贡献 (safety contribution): Safety and long-term stability constraints are embedded in the cooperative trajectory optimization framework.
- 平滑/停止避免贡献 (smoothing contribution): Optimizes throughput and continuous flow, reducing delay compared with conservative coordination.
- 创新力度评判 (reviewer judgement): 较强应用创新: 将 TD3 用于统一协同轨迹优化并强调毫秒级推理, 但可验证安全仍是短板。

1.4 核心概念对照 / Core Concepts

中文概念	English Concept	Role / 学术作用
自动交叉口管理	Autonomous Intersection Management	把信号控制替换为车辆级调度与控制。
轨迹平滑	Trajectory Smoothing	减少急加减速、停车和能耗峰值。
安全约束	Safety Constraints	把碰撞风险写成距离、时间间隔或屏障函数。
TD3 deep reinforcement learning for coordination	TD3 深度强化学习协同控制	该论文的核心方法族。

2 教材式学习路径与关键截图 / Textbook-Style Study Path & PDF Snapshots

2.1 学习路径 / Study Path

- 第一遍先读首页截图、摘要与引言, 确认研究问题和场景假设。
- 第二遍沿着技术路线图复现模型变量、约束和求解器输入输出。
- 第三遍逐节阅读 section deep reading, 对照 PDF 截图检查公式、图表和实验。
- 第四遍只站在审稿人视角重读实验, 判断 baseline、公平性、消融和鲁棒性。
- 最后把博士 checklist 写成自己的复现实验或论文拓展计划。

2.2 博士阅读 Checklist / Doctoral Reading Checklist

- 能否独立重写核心公式: MDP 目标与 TD3 策略优化 / MDP objective and TD3 policy optimization, 并解释每个符号的物理意义?
- 能否画出输入-模型-算法-控制-验证的数据流图?
- 能否指出至少两个强假设, 并说明它们在真实交通中何时失效?
- 能否复述实验基准是否公平, 指标是否覆盖效率、安全、能耗和计算时间?
- 能否提出一个与自己 JSSP/RL/smoother 课题直接相连的拓展实验?

2.3 关键截图讲解 / PDF Snapshot Explanation

Screenshots are selected anchors from the local PDFs for study and parallel reading. They do not replace the original paper.

arXiv:2205.01278v2 [eess.SY] 22 Mar 2023

Real-time Cooperative Vehicle Coordination at Unsignalized Road Intersections

Jiping Luo, Tingting Zhang, *Member, IEEE*, Rui Hao, Donglin Li, Chunsheng Chen, Zhenyu Na, *Member, IEEE*, and Qinyu Zhang, *Senior Member, IEEE*

Abstract—Cooperative coordination at unsignalized road intersections, which aims to improve the driving safety and traffic throughput for connected and automated vehicles (CAVs), has attracted increasing interests in recent years. However, most existing investigations either suffer from computational complexity or cannot harness the full potential of the road infrastructure. To this end, we first present a dedicated intersection coordination framework, where the involved vehicles hand over their control authorities and follow instructions from a centralized coordinator. Then a unified cooperative trajectory planning problem will be formulated to maximize the traffic throughput while ensuring driving safety. To address the key computational challenges in the real-world deployment, we reformulate this non-convex sequential decision-making problem into a model-free Markov Decision Process (MDP) and tackle it by devising a Twin Delayed Deep Deterministic Policy Gradient (TD3)-based strategy in the deep reinforcement learning (DRL) framework. Simulation and practical experiments show that the proposed strategy could achieve near-optimal performance in sub-static coordination scenarios and significantly improve the traffic throughput in the realistic continuous traffic flow. The most remarkable advantage is that our strategy could reduce the time complexity of computation to milliseconds, and is shown scalable when the road lanes increase.

Index Terms—connected and automated vehicles (CAVs), intersection coordination, deep reinforcement learning (DRL)

I. INTRODUCTION

WITH the proliferation of traffic demand and vehicle ownership, the current transportation systems are now facing some serious social problems: traffic congestion, accidents, fuel consumption, air pollution, etc [2]–[6]. Road intersections where multiple roads cross and merge are the main bottlenecks for urban traffic. As reported in [7], [8], congestion and accidents in these areas cause substantial economic loss and tremendous safety hazards to society, and excessive travel time for drivers. Therefore, it is critical to develop dedicated intersection management systems, to allow vehicles to cross the intersection safely and efficiently.

Traditional traffic signal control (TSC) systems, where the traffic signal timing is predetermined according to historical traffic data, is widely used in current intersection coordination systems. However, this solution may not accommodate real-time traffic flow and bursty traffic demands [9]. Benefiting from the artificial intelligence, especially reinforcement learning (RL), the adaptive TSC which could adjust traffic signal timing according to real-time traffic demand, has been shown effectiveness in reducing traffic congestion [9]–[14]. However, this paradigm is still far from satisfactory with respect to both driving safety and traffic efficiency as over 90 percent of

crashes are tied to improper human behaviors [8]. Moreover, the inconveniences of frequent stops and idling at intersections not only waste both fuel and time but also degrade the comfort of drivers and passengers [15]–[17].

In recent years, centralized coordination at unsignalized intersections has been widely studied to improve the performance of intersection control, especially when interoperations are required among vehicles. In this paradigm, there usually exists a coordination node that is fulfilled at the road side unit (RSU) to collect state information such as position, velocity, acceleration, heading angle, and intention of all involved vehicles via vehicle-to-infrastructure (V2I) communication links. After proper coordination algorithms, the maneuver instructions could be generated and forwarded to vehicles. Recent investigations show that centralized coordinators are advantageous in traffic throughput, collision-free, deadlock-free, passenger comfort, fuel efficiency, etc [15]–[19], making them suitable for real-world deployments. However, the design of centralized coordination algorithms confronts challenges due to the collision-free constraints and the complexity of the required maneuvers.

Lee and Park [20] proposed a cooperative vehicle control strategy for connected and automated vehicles to reduce stop delay, total passing time, and fuel consumption. They adopted the following two strict assumptions: (1) Any two vehicles with overlapped paths should not exist at the intersection area at the same time, thereby eliminating potential collisions coming from all conflicting approaches. (2) The acceleration of each vehicle is fixed from its initial position to the end of the intersection. Though this algorithm could improve coordination performance compared with traditional TSC, it is apparent that the above two assumptions would impose limitations on the maximum coordination capacities of the intersection. Moreover, the formulated nonlinear constrained programming (NCP) problem was solved by employing search-based algorithms such as the Active Set Method (ASM), Interior Point Method (IPM) and genetic algorithm (GA) that could produce undesirable solutions and computation time.

The authors in [21]–[26] adopted similar rules that the whole intersection area was reserved for conflicting vehicles one after another, namely *collision set* (or *critical set*) strategy. The most notable difference is that the acceleration is treated as a continuous variable rather than a fixed value for the subsequent time slots, which could further improve the coordination efficiency and passenger comfort. However, due to the non-convexity of the formulated problems, some relaxations such as the mixed integer programming (MIP) solutions are

图 1: 首页与摘要 / Title and abstract page (p.1, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Shows why learned policies are attractive for online intersection coordination but need a structured safety/smoothing interface.

required, which inevitably increase computational complexity. In [25], [26], the stability of the coordination system was also considered as a metric to trade off long-term against short-term performances. Aiming at the computational complexity, appropriate task offloading from the centralized coordinator was shown to be an option [27].

To address the underutilization issue of the intersection spatial resources, Kamal et al. [28] defined the cross-collision points (CCPs) inside the intersection area where the paths of any two conflicting vehicles intersect. The circle (with a certain radius) centering the CCP shows an approximate area where two vehicles are not allowed to enter at the same time. In addition, smooth traffic flows were achieved by solving an NCP problem at each time step over a finite horizon in the model predictive control (MPC) framework. However, as stated in this paper, MPC is usually computationally demanding and the average computation time is about 1.76 seconds per iteration, which is impractical in real-world deployments. Similarly, the intersection area can also be divided into multiple rectangular sub-zones (each could be reserved for one vehicle at a time) according to the combination of each pair of orthogonal lanes [15], [29], [30]. However, this reservation rule depends heavily on the road structure and requires much more safety redundancy than vehicle geometry.

More recently, the authors in [31] and [32] brought totally different insights that the cooperative trajectories can be optimized in the three-dimensional space with X, Y and T domains to harness the full potential of the intersection. In [32], a space-time resource searching (STRS) algorithm was proposed to search for compact and disjoint trajectories in the XYT domains, which allows multiple vehicles to share the intersection simultaneously and safely without any reservation rules, thereby providing tremendous coordination efficiency advantages over the aforementioned rule-based strategies. However, the STRS strategy shows obvious disadvantages in real-time processing performance due to the high dimensional searching space and the combination of dynamic programming (DP) and quadratic programming (QP).

In this paper, comprehensively considering the key issues, i.e., traffic throughput, driving safety, and computational complexity, at unsignalized intersections, a Twin Delayed Deep Deterministic Policy Gradient (TD3)-based centralized coordination strategy is designed in the deep reinforcement learning (DRL) framework. The contributions of this paper can be summarized as follows.

- Leveraging *Lyapunov Optimization Theorem*, a unified cooperative trajectory planning problem is formulated to maximize traffic throughput while ensuring driving safety. To harness the full coordination potential of the road intersections, we adopt the paradigm that all controlled vehicles could occupy the intersection simultaneously without any reservation rules.
- To rapidly solve the formulated non-convex sequential decision problem and obtain optimal disjoint trajectories in XYT domains, we transform this problem into a model-free Markov Decision Process (MDP) and elucidate the design of the state space, action space and reward function in detail. Then a TD3-based centralized

coordinator is trained offline and can be deployed online to meet the stringent real-time coordination requirements.

- Simulation results and laboratorial experiments demonstrate that our TD3-based strategy could achieve near-optimal performance in static coordination scenarios and significantly improve traffic throughput in continuous traffic flow. The most impressive advantage is that our strategy has millisecond computational latency, which outperforms most existing centralized strategies by which the computational complexity increases exponentially with the number of vehicles and lanes. Moreover, our strategy shows good adaptability in low and medium traffic loads compared with traditional TSC systems.

The rest of this paper is organized as follows. First, the system model is presented in Section II, including the centralized coordination framework, formulation of optimal coordination problem, and benchmark solutions. In Section III, the formulated optimization problem is transformed and solved by devising a TD3-based strategy. Section IV presents extensive simulation results and experiments to confirm the capability and applicability of the proposed approach. Finally, the conclusion and future work are presented in Section V.

II. SYSTEM MODEL

A. Intersection Coordination Framework

We consider a typical unsignalized intersection which consists of R roads (usually we set $R = 4$), each with K lanes, as shown in Fig. 1. The area within the circle is called the Centralized Control Zone (CCZ) where a centralized coordinator is deployed at the road side unit (RSU) to collect the state information of the vehicles and give instructions. The blue area is called the Conflict Area (CA) where a potential collision may occur.

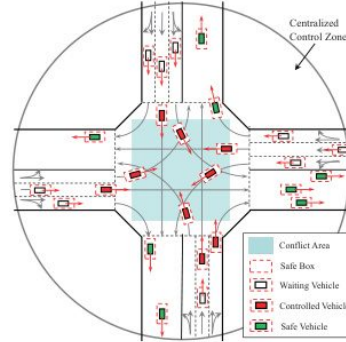


Fig. 1. Graphic model of the intersection coordination system.

In the CCZ, all vehicles will go through three different phases, i.e., waiting vehicle (WV), controlled vehicle (CV), and safe vehicle (SV). Once a vehicle enters the CCZ, it will travel at a low speed v_{low} and wait for coordination, namely WV. The WVs need to keep a safe distance from the

图 2: 建模或问题设定页 / Modeling or problem-setup page (p.2, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Shows why learned policies are attractive for online intersection coordination but need a structured safety/smoothing interface.

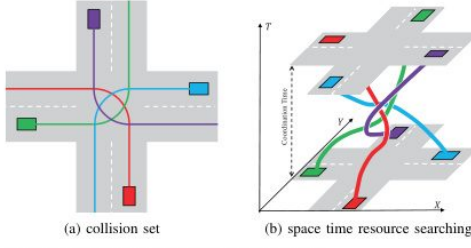


Fig. 4. An illustration of two benchmark strategies.

2) *Space-Time Resource Searching*: The space-time resource searching (STRS) strategy [32] adopts the aggressive idea that all CVs simultaneously share the intersection to improve traffic throughput. Specifically, the space and time resources in the intersection can be regarded as three-dimensional coordinates with X , Y , and T domains, as shown in Fig. 4(b). The proposition of optimizing speed profiles to make CVs travel safely on the overlapped paths in the XY domain is equivalent to searching for disjoint trajectories in XYT domains. Compared with the CS strategy, the STRS strategy shows obvious throughput advantages. By traversing space-time resources, the STRS strategy could obtain optimal passing orders and trajectories, however, at the expense of real-time processing capabilities. In practical implementations, the coordinator requires tremendous computational resources and power supply to deal with high computational complexity. Excessive problem-solving time could also lead to safety issues, especially in the CA. Moreover, the computational complexity grows exponentially with the number of vehicles and lanes, which will hinder this strategy for multi-lane scenarios.

III. DRL-BASED CENTRALIZED COORDINATOR

It is critical to rapidly obtain optimal trajectories for given initial configurations in the considered intersection coordination scenario. However, searching for optimal trajectories in 3D space XYT , or optimizing the controls on given overlapped 2D paths, takes overlong problem-solving time. Moreover, the computational complexity of the centralized coordination strategy grows exponentially with the number of lanes and the number of CVs. Thus, it is challenging to adopt the pure optimization approach-based schemes to solve the optimal coordination problem in real-time.

To address the key computational challenges, an RL approach is leveraged in this paper. We first model the sequential decision-making problem $\mathcal{P}$ as a model-free MDP and then resort to DRL algorithms to tackle it.

A. Problem Transformation

We first formulate the intersection coordination problem using the reinforcement learning (RL) framework. MDPs can be used to express the standard mathematical formalism of RL [39]. An MDP is a 5-tuple (S, A, R, P, γ) that describes the interactions between the states of the environment and

the actions of the RL agent. Specifically, at each time slot t , based on the observation $s_t \in S$, an action $a_t \in A$ sampled from a stochastic policy $\pi(a_t|s_t)$ or generated by a deterministic policy $\pi(s_t)$ is executed by the RL agent and the environment transitions to next state $s_{t+1} \in S$ with a transition probability $\Pr(s_{t+1}|s_t, a_t) \in P$, receiving a reward $r_t(s_t, a_t) \in R$. $\gamma \in [0, 1)$ is the discount factor that trades off long-term against short-term rewards. According to whether the state transition probability and the expected return from any state-action pair are prior or not, RL algorithms can be classified into two categories: model-based and model-free. In the considered intersection coordination scenario, the agent has no idea how to make a proper decision for the subsequent time slot according to the current observed environment state. Hence, it can be regarded as a model-free one.

Deep RL (DRL), which combines RL with deep neural networks (DNNs), can be applied to deal with continuous control problems. By using DNNs as a non-linear function approximator over high-dimensional state spaces, only the network parameters need to be stored instead of a lookup table that records the values of all possible state-action pairs [40]. Moreover, DNNs can infer actions on unseen observations, enabling the algorithm to obtain expected performance without traversing all state-action pairs.

1) *Observation and Action Spaces*: Intuitively, the state of each CV should contain the position and velocity information. However, this information only portrays the current driving state and ignores the intention and association among CVs. For example, if the paths of the CVs do not overlap in the CA, they could leave the CCZ as quickly as possible. Otherwise, the CVs should collaboratively cross the CA to avoid collision in the subsequent time slots.

To capture the intentions of the CVs, a natural and straightforward scheme is to assign each vehicle a tag, i.e., 0-turn left, 1-go straight, and 2-turn right. However, DNNs have difficulty in identifying the meaning of these tags. To help the DNNs understand the tags, we could encode each of these tags as a three-element binary vector, e.g., $[1, 0, 0]$ represents turning left. However, the required state dimensions to infer driving intentions increases linearly with the number of CVs, which is computationally inefficient for training the DRL agent. Finally, we use a D_r -element binary vector $\mathbf{d}_r$ to encode the joint intentions for CVs on the r -th road, where $D_r = 3^N$ if $K = 1$ else $D_r = 4^N$. This joint encode scheme shows effectiveness in identifying CVs' intentions using much fewer state dimensions.

Moreover, to capture the dynamic feature of the traffic, the queue state vector $\mathbf{I}(t)$ is also considered. We normalize the queue states to reflect the priority of different lanes, i.e.,

$$p_{r,k}(t) = \frac{I_{r,k}(t)}{\sum_{r \in \mathcal{R}} \sum_{k \in \mathcal{K}_r} I_{r,k}(t)} \quad (16)$$

Then, the environment state at time slot t can be described as:

$$\mathbf{s}_t = \{\mathbf{d}_r(t), p_{r,k}(t), x_{r,k}^n(t), y_{r,k}^n(t), v_{r,k}^n(t) \mid r \in \mathcal{R}, k \in \mathcal{K}_r, n \in \mathcal{N}_{r,k}\} \quad (17)$$

The DRL agent is trained to directly map the observation to CVs' control commands, i.e., acceleration, velocity, heading

图 3: 核心方法或算法页 / Core method or algorithm page (p.5, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Shows why learned policies are attractive for online intersection coordination but need a structured safety/smoothing interface.

Algorithm 1 Training Stage of The TD3-based Coordinator

```

1: Initialize policy parameters  $\theta$ , value parameters  $\phi_1, \phi_2$ .
   Empty replay buffer  $\mathcal{D}$ . Set target parameters equal to
   main parameters  $\theta' \leftarrow \theta, \phi'_1 \leftarrow \phi_1, \phi'_2 \leftarrow \phi_2$ . The pa-
   rameters are updated after  $T_{\text{start}}$  steps, the policy network
   and target networks are updated after  $T_{\text{delay}}$  updates of the
   critic. Initialize total time steps  $T_{\text{total}} = 0$ , total network
   updates  $T_{\text{updates}} = 0$ .
2: for  $\text{episode} = 1$  to  $E$  do
3:   Randomly initialize the position of the CVs and the
   number of WVs, receive initial observation  $s_0$ . Set time
   step  $t = 0$ , done signal  $d_0 = \text{False}$ .
4:   while not  $d_t$  do
5:     Observe current environment state  $s_t$ .
6:     if  $T_{\text{total}} < T_{\text{start}}$  then
7:       Randomly sample an action from action space.
8:     else
9:       Select action using policy  $\pi_\theta$  according to (27).
10:    end if
11:    The CVs execute control commands  $\mathbf{a}_t$  according to
    vehicle kinematic model (1).
12:    Check collision using SAT algorithm.
13:    if collide or complete then
14:      Set done signal  $d_t = \text{True}$ .
15:    end if
16:    Observe next state  $s_{t+1}$ , compute reward  $r_t$  according
    to (23), check done signal  $d_t$ .
17:    Store experience  $(s_t, \mathbf{a}_t, r_t, s_{t+1}, d_t)$  in  $\mathcal{D}$ .
18:    if  $T_{\text{total}} > T_{\text{start}}$  then
19:       $T_{\text{updates}} \leftarrow T_{\text{updates}} + 1$ 
20:      Randomly sample a mini-batch of transitions  $\mathcal{B} =$ 
       $\{(s, \mathbf{a}, r, s', d)\}$  from  $\mathcal{D}$ .
21:      Compute TD-targets according to (29).
22:      Update  $\phi_1$  and  $\phi_2$  by gradient descent using (30).
23:      if  $T_{\text{updates}} \bmod T_{\text{delay}} = 0$  then
24:        Update  $\theta$  by gradient ascent using (31).
25:        Update target networks using soft update:
           $\theta' \leftarrow \tau\theta + (1 - \tau)\theta'$ 
           $\phi'_i \leftarrow \tau\phi_i + (1 - \tau)\phi'_i$ , for  $i = 1, 2$ 
26:      end if
27:    end if
28:     $T_{\text{total}} \leftarrow T_{\text{total}} + 1$ 
29:  end while
30: end for

```

Here, the coefficients can be calculated by solving the follow-
ing linear function:

$$\begin{bmatrix} s_0 \\ v_0 \\ a_0 \\ s_1 \\ v_1 \\ a_1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 \\ 1 & T & T^2 & T^3 & T^4 & T^5 \\ 0 & 1 & 2T & 3T^2 & 4T^3 & 5T^4 \\ 0 & 0 & 2 & 6T & 12T^2 & 20T^3 \end{bmatrix} \cdot \begin{bmatrix} \omega_0 \\ \omega_1 \\ \omega_2 \\ \omega_3 \\ \omega_4 \\ \omega_5 \end{bmatrix} \quad (33)$$

As discussed in Section III-A, to cope with unbalanced
arrival rates and demands in dynamic coordination scenario,

vehicle grouping and vehicle padding methods are considered.
Specifically, WVs need to be divided into batches for the lane
with more than N WVs, and virtual vehicles will be added for
the lane with less than N WVs. For each round, the TD3 agent
is responsible for giving instructions to the CVs and ensuring
driving safety. In general, the proposed dynamic coordination
framework is summarized in **Algorithm 2**.

Algorithm 2 Dynamic Coordination Framework

```

1: The coordinator load pre-trained policy network  $\pi_\theta$ .
2: repeat
3:   Send Velocity Keeping and Safe-distance Maintaining
   commands to all WVs [47]. Divide WVs into batches
   and add virtual vehicles if needed. Current batch of CVs
   hand over their control authorities to the coordinator.
4:   Collect all state information including queue states,
   CV's driving states and intentions.
5:   Construct initial state vector according to (17).
6:   The coordinator operate in predictive mode and com-
   pute optimal trajectories for all CVs using policy  $\pi_\theta$ .
7:   Smooth the original trajectories using (32) and (33).
8:   repeat
9:     Current batch of CVs cross the intersection according
     to maneuver instructions.
10:    if CV  $(r, k, n)$  leave the CA then
11:      Turn green and keep a velocity  $v_{\text{max}}$ .
12:    end if
13:  until All CVs leave the CA (exclude virtual vehicles).
14:  Completion of current coordination. Transition to next
  batch of CVs.
15: until The coordination process is finished.

```

IV. SIMULATION RESULTS AND EXPERIMENTS

A. Simulation Settings

To demonstrate the performance of the proposed TD3-based
coordination strategy, simulation results and experiments are
presented in this section. To facilitate comparison with the
strategies described in Section II-E, we consider a special case
with batch size $N = 1$. In Section IV-D, we will further discuss
the influence of different batch sizes on the coordination
performance. The simulation procedure can be summarized
in two stages, i.e., the training stage to optimize network
parameters and the evaluation stage to test the trained agent.

In the training stage, we first train separate policies for
single-lane and multi-lane scenarios, as shown in Fig. 7 and
Fig. 1, respectively. Without loss of generality, the position,
velocity and intention of the vehicles are randomly initialized,
and the paths of the left/right-turning vehicles are line & circle
curves. The left/right-turning radius, denoted by R_l/R_r , is set
as 15(10)m/15m for single-lane case and 40m/20m for multi-
lane case. Therefore, three separate policies are trained: (1)
single-lane road intersection with $R_l = 15\text{m}$. (2) single-lane
road intersection with $R_l = 10\text{m}$. (3) multi-lane road
intersection. The training process confronts challenges due to
the high dimensionality of searching space, the complexity of

图 4: 实验或结果页 / Experiments or results page (p.8, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Shows why learned policies are attractive for online intersection coordination but need a structured safety/smoothing interface.

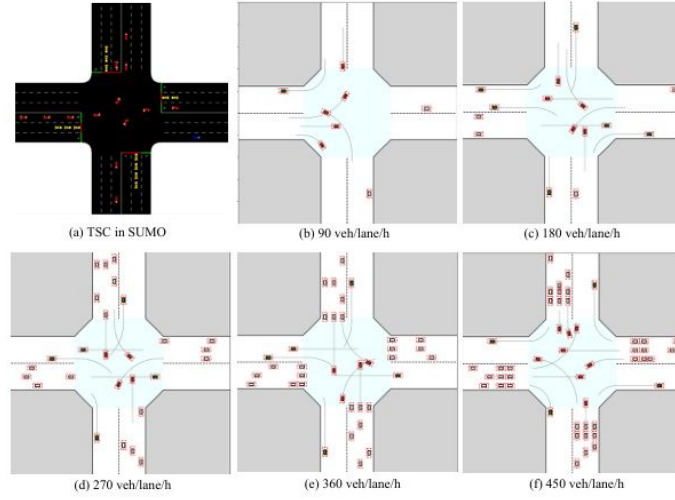


Fig. 17. Simulation platforms: (a) Traditional TSC system established in SUMO. (b)-(f) Captures of cooperative coordination animation under different traffic loads with batch size $N = 1$ implemented in Python.

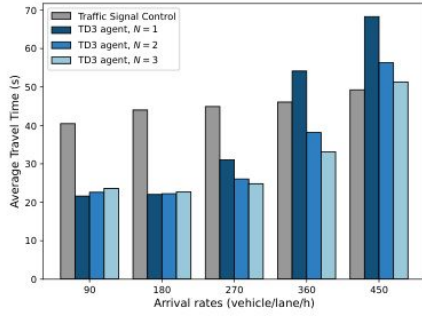


Fig. 18. Comparison of average travel time under TSC and TD3 strategies.

E. Laboratorial Experiments

Finally, we conduct experiments with four autonomous mobile robots based on the robot operating system (ROS) [48] to verify the feasibility and practicability of the proposed coordination strategy. As shown in Fig. 19, we use a laptop as the centralized coordinator and broadcast control sequences to the robots via Wi-Fi links. Each robot is equipped with a motion control unit for trajectory tracking and an Intel RealSense T265 tracking camera for self-localization. The experiment parameters are set as $v_{\max} = 0.3\text{m/s}$, $a_{\max} = 0.1\text{m/s}^2$, $R_l = 1.8\text{m}$, $\Delta T = 0.5\text{s}$, $L_{\text{car}} = 0.45\text{m}$, $W_{\text{car}} = 0.38\text{m}$, $d_{\text{lon}} = 0.5\text{m}$, $d_{\text{lat}} = 0.4\text{m}$.

Fig. 20 shows an image of the experimental coordination

scenario. Our strategy enables the four left-turning robots to occupy the intersection area simultaneously, pushing the intersection coordination capacity to its limit. Fig. 21 further analyzes the deviation of the planned trajectory (solid line) and actual trajectory (dashed line). The figure shows that both the planned and actual trajectories are within the occupied road region, illustrating that the safety redundancy is well set to circumvent safety hazards caused by system imperfections. Videos of our simulations and experiments can be found online at <https://www.youtube.com/watch?v=FEwa-YuFFAk>.

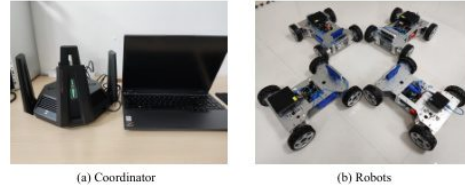


Fig. 19. Devices used in the experiments.

V. CONCLUSION

In this paper, we have investigated the cooperative vehicle coordination problem at unsignalized road intersections, where the key issues to be addressed were traffic throughput, driving safety, and computational complexity. A unified optimization problem has been formulated to maximize traffic throughput while ensuring driving safety. As the original optimization problem is computationally intractable in real-time, we have

图 5: 结论或局限页 / Conclusion or limitations page (p.13, full_page). 这张截图用于把笔记中的抽象概念拉回原论文页面, 便于你平行阅读原文、公式、图表和实验设置。与当前课题的连接: Shows why learned policies are attractive for online intersection coordination but need a structured safety/smoothing interface.

3 章节精读与逐段导读 / Section-by-Section Deep Reading & Walkthrough

p.1 I. INTRODUCTION

章节目标 / Learning goal: 建立研究动机：为什么传统信号控制、FCFS 或单车控制不足以处理该论文关注的交叉口协同问题。

Understand why the paper needs a new coordination method rather than a standard signal, FCFS, or single-vehicle controller.

关键论点 / Key argument: 这一部分通常把痛点落到 DRL cooperative trajectory optimization：既要提高效率，又要保持安全与在线可执行。

技术焦点 / Technical focus: 精读时标出作者批判的 baseline、场景假设、交通参与者类型和隐含优化目标。

审稿追问 / Reviewer question: 作者指出的痛点是否真的需要新方法，还是可以由更强的优化器/更保守规则解决？

p.2 II. SYSTEM MODEL

章节目标 / Learning goal: 把自然语言交通问题压缩为变量、状态、约束、目标函数或图结构。

Translate the traffic problem into variables, states, constraints, objectives, or graph relations.

关键论点 / Key argument: 本节是复现论文的核心入口，应与公式“MDP 目标与 TD3 策略优化 / MDP objective and TD3 policy optimization”一起读。

技术焦点 / Technical focus: 逐项检查车辆状态、冲突关系、控制输入、时间/空间索引和安全阈值是否定义完整。

审稿追问 / Reviewer question: 这些建模假设在混合交通、通信延迟、感知误差和高密度车流下是否仍成立？

p.3 3. The road region occupied by the vehicle is defined by a two-

章节目标 / Learning goal: 承接前后章节，补充局部定义、案例或实现细节。

Recover implementation details needed for reproduction.

关键论点 / Key argument: 这类章节常常不是主贡献，但决定你能否真正复现方法。

技术焦点 / Technical focus: 检查它是否给出了参数、伪代码、场景划分或实现细节。

审稿追问 / Reviewer question: 跳过该节会不会导致公式或实验无法复现？

p.5 III. DRL- BASED CENTRALIZED COORDINATOR

章节目标 / Learning goal: 解释算法如何把模型转化为可执行决策，并说明在线运行机制。

Trace how the paper turns the model into executable online decisions.

关键论点 / Key argument: 将协同控制表述为连续动作 MDP，TD3 使用双评论家取最小值、延迟 actor 更新和目标策略平滑来提升稳定性。在线执行时只需前向传播策略网络，因而推理很快。

技术焦点 / Technical focus: 画出输入、核心求解器、输出和安全检查接口，明确哪些步骤是优化、哪些是规则、哪些是学习。

审稿追问 / Reviewer question: 若算法某一步失败，论文有没有 fallback、重规划或安全过滤机制？

p.8 1 T T 2 T 3 T 4 T 5

章节目标 / Learning goal: 承接前后章节，补充局部定义、案例或实现细节。

Recover implementation details needed for reproduction.

关键论点 / Key argument: 这类章节常常不是主贡献，但决定你能否真正复现方法。

技术焦点 / Technical focus: 检查它是否给出了参数、伪代码、场景划分或实现细节。

审稿追问 / Reviewer question: 跳过该节会不会导致公式或实验无法复现？

p.8 IV. SIMULATION RESULTS AND EXPERIMENTS

章节目标 / Learning goal: 验证方法是否在公平基准下改善效率、安全、能耗或计算时间。

Check whether the claimed improvements are supported by fair baselines and meaningful metrics.

关键论点 / Key argument: 实验应与论文声称的贡献闭环：Simulation and practical experiments; authors report millisecond-scale computation and scalability with lane count.

技术焦点 / Technical focus: 重点追踪指标：平均通行时间、吞吐量、延误、速度波动、碰撞/冲突次数和在线计算耗时。同时检查仿真平台、交通需求和随机性设置。

审稿追问 / Reviewer question: 实验是否只展示平均收益，还是覆盖了高密度、异常扰动和失败案例？

p.13 V. CONCLUSION

章节目标 / Learning goal: 提炼作者承认的边界，并把它转化为博士课题的可拓展问题。

Convert acknowledged boundaries into research opportunities.

关键论点 / Key argument: 结论不只是总结结果，更是观察作者没有解决什么的窗口。

技术焦点 / Technical focus: 把 future work 与前文假设逐一对应，寻找可发表的补强点。

审稿追问 / Reviewer question: 如果我是审稿人，我会要求补哪一个实验、定理或消融？

3.1 逐段式导读 / Paragraph-Level Walkthrough

Opening motivation / 开篇动机段 先把作者的痛点翻译成一句博士问题：能否用连续动作深度强化学习直接学习无信号交叉口的实时协同轨迹控制策略？读这一段时不要急着接受动机，要追问传统 FCFS、信号灯或保守 MPC 为什么不足。

Turn the opening motivation into one doctoral-level question and test whether the paper truly needs a new method.

Assumption and setting / 假设与场景段 把车辆类型、通信条件、路口类型和动力学假设列成表。该文的关键边界包括：训练环境覆盖了测试场景的关键交通分布。；中心协调器能获得足够完整的车辆状态。；奖励函数能充分惩罚碰撞、剧烈加减速和低效率。

List vehicle type, communication, intersection setting, and dynamics assumptions before reading the equations.

Model and notation / 模型与符号段 围绕核心公式“MDP 目标与 TD3 策略优化 / MDP objective and TD3 policy optimization”复写符号表，确认每个变量是决策变量、状态变量、参数还是约束阈值。

Rewrite the symbol table around the core formulation and classify variables as states, decisions, parameters, or constraints.

Algorithm mechanism / 算法机制段 把算法读成一条数据流：输入交通状态，执行 Formulates unified cooperative trajectory optimization for unsignalized intersections and solves the sequential decision problem with TD3., 输出可执行通行/控制决策，并检查安全接口。

Read the algorithm as a dataflow from traffic state to executable sequence/control decision, with explicit safety interfaces.

Experiment and evidence / 实验证据段 不要只看作者报告的提升，要检查基准、公平性和指标。本文验证描述为：Simulation and practical experiments; authors report millisecond-scale computation and scalability with lane count.

Go beyond reported gains and inspect baselines, fairness, metrics, and computation time.

Reviewer synthesis / 审稿式总结段 最后把贡献拆成可发表与需补强两部分。对你课题最直接的用途是：Shows why learned policies are attractive for online intersection coordination but need a structured safety/smoothing interface.

Separate publishable contribution from missing evidence, then connect the result to your own thesis architecture.

4 技术路线图与贡献量评估 / Technical Route & Contribution Matrix

Step	Stage / 阶段	Content / 内容
1	输入 / Input	车辆状态、路径/车道、交通需求、信号或协同信息。
2	建模 / Modeling	MDP 目标与 TD3 策略优化 / MDP objective and TD3 policy optimization
3	核心求解 / Core solver	Formulates unified cooperative trajectory optimization for unsignalized intersections and solves the sequential decision problem with TD3.
4	安全接口 / Safety interface	Safety and long-term stability constraints are embedded in the cooperative trajectory optimization framework.
5	平滑与效率 / Smoothing and efficiency	Optimizes throughput and continuous flow, reducing delay compared with conservative coordination.

6	验证 / Validation	Simulation and practical experiments; authors report millisecond-scale computation and scalability with lane count.	
Dimension / 维度		Score	Comment / 评语
理论贡献 / Theoretical contribution		3/5	MDP 目标与 TD3 策略优化 / MDP objective and TD3 policy optimization
算法贡献 / Algorithmic contribution		4/5	Formulates unified cooperative trajectory optimization for unsignalized intersections and solves the sequential decision problem with TD3.
实验贡献 / Experimental contribution		3/5	Simulation and practical experiments; authors report millisecond-scale computation and scalability with lane count.
工程贡献 / Engineering contribution		3/5	Safety and long-term stability constraints are embedded in the cooperative trajectory optimization framework.
博士课题价值 / PhD relevance		5/5	Shows why learned policies are attractive for online intersection coordination but need a structured safety/smoothing interface.

5 理论方法论深度解构 / Methodology & Theoretical Framework

5.1 问题数学建模 / Mathematical Formulation

MDP 目标与 TD3 策略优化 / MDP objective and TD3 policy optimization

$$\max_{\pi_{\theta}} \mathbb{E} \left[\sum_{t=0}^T \gamma^t r(s_t, a_t) \right], \quad a_t = \pi_{\theta}(s_t), \quad y = r + \gamma \min_{k=1,2} Q_{\phi_k}(s', \pi_{\bar{\theta}}(s'))$$

Symbol / 符号	含义	Meaning	Role / 建模作用
s_t	协同状态	cooperative state	包含车辆位置、速度和交互信息
a_t	连续控制动作	continuous action	通常对应加速度或轨迹控制量
r	奖励函数	reward	平衡效率、安全和舒适性
gamma	折扣因子	discount factor	决定长期收益权重

Q_phi	评论家网络	critic network	估计动作价值并抑制过估计
-------	-------	----------------	--------------

5.2 核心算法架构 / Core Algorithm Layout

将协同控制表述为连续动作 MDP，TD3 使用双评论家取最小值、延迟 actor 更新和目标策略平滑来提升稳定性。在线执行时只需前向传播策略网络，因而推理很快。

The cooperative control problem is cast as a continuous-action MDP; TD3 uses clipped double-Q learning, delayed policy updates, and target smoothing for stable training and fast inference.

5.3 收敛性、稳定性或实时性 / Convergence, Stability, Real-Time Feasibility

训练稳定性来自 TD3 机制而非形式化安全证明；实时性来自神经网络推理。若状态分布偏移，策略可能给出未认证动作。

TD3 improves learning stability, but safety remains empirical unless wrapped by a constraint layer or safety filter.

5.4 潜在假设与边界条件 / Assumptions & Boundary Conditions

- 训练环境覆盖了测试场景的关键交通分布。
- 中心协调器能获得足够完整的车辆状态。
- 奖励函数能充分惩罚碰撞、剧烈加减速和低效率。

6 实验验证与结果评判 / Experimental Validation & Result Evaluation

Baselines & Environment / 基准与环境：传统协同控制、非协同策略、FCFS/规则控制和其他深度强化学习控制器。

Validation / 验证：Simulation and practical experiments; authors report millisecond-scale computation and scalability with lane count.

Metrics / 指标：平均通行时间、吞吐量、延误、速度波动、碰撞/冲突次数和在线计算耗时。

Ablation / 消融：应检查奖励项、双评论家、目标策略平滑和安全惩罚各自贡献；若缺少这些消融，审稿时会质疑可解释性。

Reviewer reading / 审稿式阅读提醒：阅读实验时不要只看平均值；应检查交通密度、车流方向、随机种子、失败案例和计算耗时。For doctoral reading, inspect density regimes, demand patterns, random seeds, failure cases, and computation time rather than only mean improvements.

6.1 图表阅读线索 / Figure and Table Reading Cues

PDF extraction detected approximately 55 figure references and 1 table references. Exact captions remain in the original PDF; the bilingual cues below tell you what to inspect first.

图表名称	Figure/Table Name	Reading focus / 阅读重点
系统框架图	system framework diagram	先确认输入、决策层、控制层和安全约束之间的数据流。
策略网络/训练流程图	policy-network or training-pipeline diagram	检查状态、动作、奖励、经验回放和安全惩罚是否闭环一致。
实验指标对比图表	experimental metric comparison plots/tables	重点看平均值之外的密度变化、失败场景、计算时间和方差。

7 审稿人视角：批判性思考与未来方向 / Critical Thinking & Future Directions

7.1 论文局限性 / Limitations & Weaknesses

- 学习策略难以给出严格安全证书。
- 仿真到真实 (sim-to-real) 分布偏移可能显著。
- 集中式状态采集和控制权切换在工程落地中成本较高。

7.2 博士课题拓展点 / Research Extensions for PhD

- 在 TD3 actor 后接 CBF/MPC safety shield, 形成 learn-then-filter 架构。
- 把 JSSP 调度作为高层离散动作, TD3 只负责连续轨迹平滑。
- 用离线强化学习或保守 Q 学习降低在线探索风险。

7.3 如何服务当前课题 / Use in This Project

Shows why learned policies are attractive for online intersection coordination but need a structured safety/smoothing interface.

8 交互式可视化与 PDF 排版蓝图 / Interactive Visualization & PDF Layout Blueprint

动态参数 / Parameter: 安全奖励权重 / Safety reward weight, range [0.1, 5.0] .

权重提高通常降低冲突风险, 但过高会造成保守减速和吞吐下降。

Higher safety reward reduces conflicts but may make the policy conservative.

8.1 HTML 结构化布局建议 / HTML Structuring

- 顶部固定 metadata bar: 题名、作者、年份、主题、PDF 页数。
- 左侧目录/section navigator, 右侧为五大精读模块。
- 公式卡片使用 `<pre>` 或 LaTeX block, 紧跟符号表。

- 审稿人批注用 reviewer-note block，博士拓展用 research-idea list。
- 交互参数用 input[type=range] + canvas 曲线，打印 PDF 时隐藏交互控件但保留解释。

9 来源锚点与逐节阅读路径 / Source Anchors & Section-by-Section Reading Map

Page	Extracted heading
p.1	I. INTRODUCTION
p.2	II. S YSTEM MODEL
p.8	IV. SIMULATION RESULTS AND EXPERIMENTS

p.1 I. INTRODUCTION 定位问题背景、研究缺口和为什么现有保守/非协同方法不足。/ Frames the motivation, research gap, and limits of conservative or non-cooperative methods.

p.2 II. S YSTEM MODEL 把交通交互转写为状态、约束、目标或图结构，是精读时最应复现的部分。/ Defines states, constraints, objectives, or graph structures; this is the section to reproduce carefully.

p.3 3. The road region occupied by the vehicle is defined by a two- 作为局部论证段落阅读，关注它如何服务主问题。/ Read as a local argument and ask how it supports the main question.

p.5 III. DRL- BASED CENTRALIZED COORDINATOR 作为局部论证段落阅读，关注它如何服务主问题。/ Read as a local argument and ask how it supports the main question.

p.8 1 T T 2 T 3 T 4 T 5 作为局部论证段落阅读，关注它如何服务主问题。/ Read as a local argument and ask how it supports the main question.

p.8 IV. SIMULATION RESULTS AND EXPERIMENTS 检验方法是否真的改善效率、安全或能耗；要关注基准是否公平。/ Tests efficiency, safety, or energy claims; inspect whether baselines are fair.

p.13 V. CONCLUSION 总结贡献和未来方向，也常暴露作者承认的边界条件。/ Summarizes contributions and often reveals author-recognized boundaries.